

CBC Transition in Kenya: Executive Summary

Project: Data-Driven Risk Assessment for 47 Counties

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Problem Statement

Kenya’s shift from 8-4-4 to CBC has faced teacher shortages, infrastructure gaps, digital divide, and over 150,000 student dropouts. Policymakers lack evidence to target interventions. This project identifies which counties need resources first.

Key Findings

Metric	Non-ASAL Counties	ASAL Counties	Gap
Average FLANA Score (%)	65.8	56.2	9.6 points
Teacher shortage severity	Low	Severe	—
Digital access	High	Very Low	—
Transition risk	Low	High	—

Statistical significance: $p < 0.001$ (t-test)

Teacher Shortage Impact

- Moving from "Low" to "Severe" teacher shortage drops FLANA score by **7.5 percentage points**.
- Regression coefficient: -2.35 ($p < 0.001$)
- $R^2 = 0.74$ (teacher shortage + digital access explain 74% of variation)

Digital Divide

- "High" digital access counties: 68.2% average FLANA
- "Very Low" digital access counties: 56.8% average FLANA
- **Gap: 11.4 points**

County Clusters (K-Means Analysis)

- **High Performing (20 counties):** Non-ASAL, low teacher shortage, high digital access (e.g., Nyeri, Kiambu)
- **Moderate Risk (13 counties):** Mixed, moderate shortages (e.g., Kisumu, Mombasa)
- **Critical Need (14 counties):** All 11 ASAL counties + Kakamega, Bungoma, Kilifi, Siaya, Kwale, Homa Bay, Migori

Student Dropout Crisis

- Grade 3 cohort (2019): 1,282,150 learners
- Grade 9 cohort (2025): 1,130,587 learners
- **Lost: 151,563 students (11.9%)**

Recommendations

1. **Target 14 Critical Need counties first** – 70% of new teacher recruitment and digital infrastructure.
2. **Hire 200+ STEM teachers per Critical Need County** within 12 months.
3. **Provide solar-powered computer labs** to all "Very Low" digital access counties.
4. **Launch a public CBC Transition Risk Dashboard** at the Ministry of Education.

Tools Used

- **Tier 1:** Google Sheets (pivot tables, descriptive stats, risk scoring)
- **Tier 2:** Power BI (interactive dashboard with map, slicers, drill-through)
- **Tier 3:** Python (pandas, seaborn, scikit-learn, SciPy) – correlation, clustering, regression, t-test

Value to employer

This project demonstrates end-to-end analysis: problem definition, data gathering, cleaning, exploratory analysis, interactive dashboarding, statistical modelling, and actionable recommendations. I can turn real-world chaos into data-driven strategy.